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Unsupervised feature selection and class labeling for credit card fraud

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Abstract

Large datasets frequently lack class labels, and obtaining labeled data often involves substantial financial and time costs, along with risks of label noise and inaccuracies due to manual annotation. In the context of fraud detection, such as credit card fraud, these challenges are compounded by privacy concerns and high class imbalances, which severely degrades classification performance of machine learning models. In this paper, we present a fully unsupervised approach that combines SHapley Additive exPlanations (SHAP) for feature selection with an autoencoder based method for generating class labels for a widely used credit card fraud detection dataset. Using this publicly available and well-known dataset, we construct different sized datasets using feature selection, generate class labels, and measure the quality and efficacy of the labels. We evaluate the labels by training different types of supervised classifiers on the newly generated labels and measure their Area Under the Precision-Recall Curve (AUPRC). Empirical results show that using SHAP feature selection consistently and significantly improves the quality and usability of the generated class labels, as measured by the AUPRC performance of classifiers trained on them. Results also show that the generated labels, both with and without a feature selection preprocessing step, outperform Isolation Forest (IF), an unsupervised anomaly detection method used as a baseline. This demonstrates that SHAP-based feature ranking and selection significantly improves generated class label quality for credit card fraud detection and is a promising strategy for handling large, imbalanced, and unlabeled fraud detection datasets.

Keywords: Label generation, Unsupervised learning, Credit card fraud detection, SHAP, Feature selection

Introduction

Obtaining quality labels for classification modeling is a crucial task for machine learning. The labeling process is often a resource-intensive and costly endeavor which is prone to errors and inconsistences [1]. The quality of the labels is an important factor for high performing machine learning models as noisy or inaccurately labeled data can severely impact the effectiveness and reliability of classification models [2]. To add to these challenges, a significant portion of newly created datasets are unlabeled by default [3]. This abundance of unlabeled data presents both an opportunity and challenge for machine learning. While machine learning models often do better with more data [4],

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the scarcity of class labels in domains such as medical image diagnosis or fraud detection significantly limits the viability of supervised models, which require labels to function. These domains face additional challenges related to privacy concerns since labeling often requires the need for human experts to manually label each row [5]. Privacy-related concerns also limit the public availability of financial datasets, such as credit card transactions datasets used for fraud detection. An unsupervised and automated approach can handle the data labeling process. Since human interaction is not required privacy can be preserved. These types of models directly learn from the dataset's features and algorithmic labeling alleviates the privacy concerns by removing direct human involvement. However, while unsupervised learners can be used for classification tasks, such as in fraud detection [6], they often underperform compared to their supervised counterparts [7], in general. This highlights the need for advancements in machine learning research to address the challenges associated with unlabeled data and creating high quality class labels.

Feature selection is a crucial data reduction technique that addresses several problems inherent in machine learning, especially in the context of large and imbalanced data, as evident in fraud data. By identifying and retaining only the most relevant and important features, feature selection techniques reduce the dimensionality of the data, which in turn simplifies the model. A simplified model has reduced computational costs, is easier to interpret, and simplification of the model often improves model performance [8, 9]. This is particularly important in scenarios with high class imbalance. Class imbalance, with respect to the binary classification problem, is when the class representation of one class is significantly skewed compared to the other. A dataset can be considered imbalanced when the ratio of one class to the other exceeds 1:4 (minority:majority ratio). Severe, or high class imbalance is when the ratio shrinks and can surpass 1:1000 [10]. The issues that arise from class imbalance are only further exaggerated as the imbalance ratio increases and special considerations for effective model training and classification are needed [11]. Our work aims to use feature selection to improve the quality of labels generated by an existing methodology [12] specifically designed to address the challenges of unlabeled data and high class imbalance, by generating new class labels using only the most important features, in an unsupervised way.

Typically, the feature selection process is supervised. Our work focuses on the unsupervised and unlabeled scenario. Thus, an unsupervised feature selection technique is needed. SHapley Additive exPlanations (SHAP) [13] offers a robust method for feature selection by providing a ranking of feature importance that can be applied regardless of whether the data is labeled or unlabeled. For the unlabeled scenario, which we focus on in this work, first an unsupervised learner is trained on the dataset, then the SHAP kernel explainer function and the SHAP values function are used to rank the feature by importance. Importantly, the class label is neither required nor used when using SHAP. While supervised feature selection methods have shown promise, our motivation for an unsupervised SHAP approach is that it aligns with both our unsupervised methodology and many real-world scenarios where data is unlabeled. By leveraging the output of an unsupervised model, Isolation Forest, this approach provides a viable way to perform feature selection when labels are unavailable. To evaluate whether SHAP feature selection improves generated label quality, we apply it to the widely used and publicly

available Credit Card Fraud Detection dataset [14, 15], which is the only publicly available real-world datasets of its kind.

We use SHAP, in an unsupervised manner, to perform feature selection and create two new datasets that use only a subset of the original features. We then apply the unsupervised labeling technique to each of the datasets and measure the quality of the labels with and without feature selection. We provide empirical results measuring the label quality and compare against an unsupervised baseline, namely Isolation Forest (IF) [16].

Using machine learning techniques to analyze such data is critical for several reasons. The scale of credit card fraud is immense, with billions of dollars lost globally each year, which places a significant financial strain on the economy and leads to increased costs for all credit card users [17]. Advances in machine learning can help reduce these losses by enabling quicker and more accurate detection of fraudulent credit card use. Additionally, machine learning offers speed advantages and can uncover patterns that might be missed by manual inspection. To the best of our knowledge, this is the first work to use unsupervised feature selection in conjunction with an unsupervised label generating method for labeling credit card fraud detection data, making this work a meaningful contribution to the field of machine learning research.

The remainder of this paper is organized as follows. Section Related works presents related works in the context of automatic data labeling, feature selection and highlights how this research is novel. Section Methodology details the unsupervised feature selection and how we use it in conjunction with the unsupervised label generating technique. In Sect. Experimental design we provide detail on the credit card datasets, how we measure new label quality and performance. Section Results and analysis presents our empirical results that show the benefits of feature selection and the efficacy of the overall work. Section Conclusion concludes the work and presents potential areas of future work.

Related works

Hancock et al. [18] detail how Shapley Additive exPlanations (SHAP) can be utilized as an effective tool for feature selection in the binary classification scenarios where the dataset has both class labels, only labels from one class, and the unlabeled scenario. They show that SHAP-based feature selection is a versatile technique and is a valuable method for data reduction without compromising model performance. Their study uses credit card fraud datasets both with and without class labels. However, their work using SHAP in the unlabeled scenario is what is most applicable to our work. When working without class labels the authors explore the use of an Isolation Forest in combination with SHAP feature selection. Their results show that SHAP values can be used to assign importance to features, even without the use of class labels and is able to be applied to a dataset where the input features are anonymous and derived from PCA analysis from unknown sources. Though their research spans the three different labeled scenarios, their unsupervised SHAP research is applicable to ours. We build upon their work by utilizing SHAP, in an unsupervised manner, as a feature selection technique. By ranking the features and selecting the most important features, we are able to reduce the number of features without the need for class labels, and then are able to measure its ability to improve upon generated class labels for credit card fraud detection.

Salekshahrezaee et al. [19] present a related work that also uses the same dataset as our paper. The authors explore data sampling techniques, such as Random Undersampling (RUS), which aims to reduce the size of the dataset. By randomly sampling from the majority class, RUS aims alleviate the issues of class imbalance by producing a dataset with a more balanced class ratio. RUS requires the use of class labels. They also use feature extraction techniques, specifically PCA and a Convolutional Autoencoder (CAE) [20], both unsupervised approaches. In contrast, we utilize SHAP for feature selection, which also reduces the dataset size but does not change the class ratio, nor does it need class labels to rank and select the best features. SHAP also does not extract features, where new features are created by combining features, from the datasets but rather selects from the existing ones. Another potential shortcoming is that the authors applied PCA to a dataset that has features created from the output of PCA. Though both have the same effect of reducing the computational requirements of a model trained on a fewer number of features. An additional, and important, distinction is our work uses the SHAP feature selection method to improve the quality of newly created class labels, not directly to improve classifier performance that uses the existing labels.

In [21], Jiang et al. propose a new unsupervised learning method to detect fraudulent transactions. They highlight how traditional machine learning methods, such as support vector machines (SVM), Random Forest, and Decision Trees, often struggle with class imbalance and with unknown fraud patterns. Their proposed algorithm, Unsupervised Attentional Anomaly Detection Network uses autoencoders with feature attention and Generative Adversarial Networks (GANs) [22]. Their results show that their method outperforms traditional machine learning methods and some existing deep learning models by better handling class imbalance. They use the same credit card fraud detection dataset we use in this work. Their work is related to ours in that their work focuses on unsupervised methods and uses a method for determining feature importance. However, their model's feature attention, a channel-wise feature attention module, does not specifically perform feature selection, where less important features are removed from the dataset. Our work uses an unsupervised feature selection method that aims to reduce the number of features to improve performance.

In [23], the authors evaluate the performances of three techniques for classifying the credit card fraud dataset, as used in our work and other previous related works. Their work is similar to ours in that they focus on techniques that are trained in an unsupervised manner. They use a one-class SVM [24], Autoencoder [25], and the Robust Mahalanobis technique. The first two are well-known machine learning techniques with open-source implementations, the latter is an outlier detection method that uses the Mahalanobis distance metric. However, their work differs from ours in that they do not use any feature selection techniques, and the unsupervised anomaly detection method used in our work serves only as a baseline when measuring the quality of our generated labels. Another important distinction is that though they train in an unsupervised setting, class labels are still required. Training a one-class SVM requires examples from only one class. This means that their approach cannot be considered entirely unsupervised since labels are required to prepare the training data for the one-class SVM. Our work is set apart since we only use class labels for performance comparisons, and our work is entirely unsupervised.

Moslehi et al. [26] adopt a clustering approach combined with an algorithm designed to label the clusters. They use K-means clustering to improve upon the clustering process while operating under the premise that labeling the entire dataset may not be feasible. Instead, they propose to label a smaller portion of the data based on clusters formed within the same subset. Their goal is to improve the quality of labels assigned to the clusters. To do this, they use a secondary dataset for the source of their labels and generate statistics of those instances using their clustering techniques. These calculated statistics are then matched with similarly calculated statistics from another cluster and the cluster with the most similar statistics provides the label, i.e., they have two sets of clusters, ones with labels and ones without. If the statistics calculated on the labeled cluster are similar to those calculated on the unlabeled cluster, that label is assigned to the unlabeled cluster. Their experimental work shows that this is an improvement over techniques documented in previously published research. Similar to this, Rauber et al. [27] propose a method to label self-organizing maps [28] applied to a small text dataset. In this work, they assign labels to the self-organizing map clusters that are pulled from a set of words that are in the features of the dataset. Maqbool et al. [29] present an approach for labeling clusters by ranking keywords using frequency and inverse frequency, their approach assigns labels to clusters. Their work is based on keyword identifiers that are in their dataset. Our work is automated in nature, like [26] and [29], but it differs from these works in that we do not use the features directly as part of the labeling nor do we start with a secondary labeled dataset from which labels originate.

These related works differ from ours in a few significant ways. First, the unsupervised feature extraction research is used in a distinctly different way than how we use feature selection. In [19], the authors extract features rather than rank and use a subset of them. Further, our work aims to improve upon the quality of generated class labels, not supervised learning of a labeled dataset. Another important difference is that in some of the related works that use clustering approaches, significantly smaller datasets were used. These datasets also do not have anonymized features. In one case, the labels used in cluster labeling were found in the features. In another case, the authors required a secondary and supplementary labeled dataset. Our work is entirely unsupervised, the labels provided by the dataset we use are not used in the feature selection process nor the labeling generation process. Additionally, we aim to build upon the work in [12] to apply feature selection to our datasets, in an unsupervised manner, before generating class labels in an unsupervised fashion. After which, we can measure how SHAP affects the quality of labels generated by our method.

Methodology

The methodology used in this paper was designed for generating binary class labels for highly imbalanced large datasets [12] and the application of our method to balanced datasets is outside the scope of this paper. In combination, we apply SHAP feature selection to try and improve upon the quality of generated class labels. To the best of our knowledge, this is the first work of its kind in this area.

The first step in our method is to apply SHAP to the input dataset in an unsupervised way. One significant advantage is that to effectively apply SHAP, expert knowledge of the dataset's attributes is not required [18]. We do not have much information about our

dataset's features, since they were anonymized with PCA, more details regarding this dataset are in Sect. Experimental Dataset.

SHAP [13] is a method that can be used to explain machine learning models. The Shapley value, based on game theory, is an importance score assigned to each feature of a model by considering how its presence and absence affects a model's predictive outcome. SHAP can be used to provide both global and local interpretability. The global explanation provides an overall view of the model's behavior across an entire dataset. For example, which features are the most important across all of the given predictions. The local explanation provides insight into why a model made a specific prediction. That is, it provides an explanation on how each feature contributed to a particular prediction. A combination of global and local explanations was used in the feature ranking used in this study by using a subset of the data for calculating SHAP values. First, an unsupervised Isolation Forest was trained on the entire dataset using all the dataset's features. Next the SHAP kernel explainer is applied to the trained model in conjunction with a subsample of the dataset. Then the "shap_values" function is used, being passed a different subset of the data, to create a list of attributes and their SHAP values that show how much each of the attribute contributes to the classifier. This function is individually applied to each instance in the given data subset. Since these values can be negative or positive, we use the absolute value of the SHAP values to measure the magnitude of each value. The average SHAP value, for each of the attributes of the data, is used as the measure of feature importance.

Once the feature importance is determined by SHAP for all features, the most important features are chosen according to their mean absolute SHAP value. The top 3, 5, 7, 10, 15, and 29 (all) features were evaluated [18]. In their work, an IF model was trained using all six levels of feature selection and classification performance metrics were calculated across 50 replications each, for a total of 300 experimental trials. Analysis of variance (ANOVA) was performed, and the results were ranked using Tukey's Honestly Significant Difference (HSD) test. The results showed that the top 15 and 29 features produced statistically similar feature selection results, followed by using the top 10 features. We chose to use the three sets that yielded the highest performance. Thus, for our work we use the top 10 features and the top 15 features, as determined by SHAP, as well as all 29 features.

The labeling approach involves using an Autoencoder [25] as the underlying unsupervised learner that effectively learns from only the datasets features. An error metric is created for each instance after training and then the instances are ranked from highest error to lowest error. This reconstruction error calculates the differences between the input features and the features that are reconstructed by decoder portion of the autoencoder, discussed in more detail below. Thus, the error metric used is solely based on dataset features. Instances that have a higher error are more likely to belong to the fraudulent class while the ones with lower error are more likely to be genuine or non-fraud transactions. The list of instances ranked by error results in a gradient from most likely fraudulent to least likely fraudulent. A threshold is established, above which instances are labeled as fraud and genuine otherwise. To enhance the minority class, a portion of the instances that fall just below the threshold are reclassified as fraudulent, or the minority class. The total number of instances labeled

positive, or fraudulent, is denoted as P, and can be set by the user. Since the instances nearest to the threshold are not strongly associated to the majority class, including them in the positive class aims to increase diversity within the minority group, thereby enhancing model generalization and performance. As P increases, we intend to produce better quality labels, and the classifiers trained on them achieve higher classification performance metrics. These boundary instances could be considered noisy, and the alternative would be to remove these instances from the dataset. However, removing them might result in the loss of potentially valuable information.

We use a fully connected autoencoder, a type of artificial neural network. They are typically used for unsupervised learning and are made up of an encoder and decoder section. Autoencoders are a good fit for our type of dataset since they are an effective learner in the presence of class imbalance [30–32]. Generally, during training, the encoder learns to compress the input into a lower-dimensional representation, while the decoder reconstructs this representation back to the original input dimensionality. However, depending on the number of input features, the autoencoder may initially perform an expansion from the input dimension to the number of neurons in the first hidden layer before undergoing the compression and subsequent reconstruction process.

The autoencoder uses a reconstruction error to quantify the difference between the input and output data. This is the error we use to sort our instances, once the network is trained. Specifically, we use mean squared error (MSE) as the error function. MSE calculates the average of the squares of errors between the actual and predicted values. MSE is defined as $\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2$, where n is the total number of data points, x_i is the i-th element of the input data vector x, and $\hat{x}_i$ is the i-th element of the reconstructed output vector $\hat{x}$ produced by the autoencoder.

In general, autoencoders are like many other neural networks, where the weights between neurons are adjusted incrementally to minimize an objective function. In our case, we train the autoencoder using backpropagation. We use a fully connected autoencoder where the input layer contains the same number of neurons as there are features in the dataset it is applied to. This input layer would have a size of 29, 15, or 10, depending on the number of features in the dataset it is generating labels for. The encoder section starts with the input size of the number of features, then a 100-neuron layer fully connected to a second hidden layer with 50 neurons. Both use the ReLU activation function. The decoder section is a reversal of this but uses the Tanh activation function, while the final output layer uses ReLU. The autoencoder is trained using a learning rate of 0.0001, a batch size of 256, the Adam optimizer, and MSE as the loss function. We set 20% of the data as a validation set size and use the EarlyStopping function to monitor the training loss, set a patience of 25 epochs and a maximum of 250 epochs. The autoencoder was defined, trained, and used for predictions with Keras [33], version 2.8.0. It is important to reiterate that both the autoencoder steps and the SHAP feature selection steps do not use the class labels. This is critical because in practice, these steps would be applied to an entirely unlabeled dataset.

The output of the methodology is a dataset that has fewer features, as selected by SHAP, and class labels, as generated by the label generating method. For our

experiments, we have a dataset with the top 15 features, the top 10 features, and all features in combination with P set to 500, 1000, and 1500. Resulting in 9 separate datasets that need to have their class labels evaluated.

Experimental design

We designed our experiments to evaluate both the quality and effectiveness of our method's generated class labels as well as what the effects are of unsupervised feature selection on the quality and performance of labeling for imbalanced fraud data. To do this, we start by using SHAP to select the top 10 and 15 features for our dataset, giving us three total datasets. Then we apply the unsupervised labeling technique to each of the datasets, using only the features of the respective datasets, to generate fraud and non-fraud class labels for the dataset. After generating the class labels, we train supervised classifiers using only the new labels and assess their classification performance against the original labels, which serve as the ground truth labels. By comparing these performance results against that of an unsupervised baseline trained on the same datasets, we effectively evaluate the efficacy of the new labels with and without feature selection.

Experimental dataset

The dataset used in our work is the publicly available and widely referenced credit card fraud detection datasets, initially introduced in [14], and is accessible via [15]. This dataset was created through a collaborative effort between Worldline, a large payment processing company and the Université Libre de Bruxelles. It includes over 284,000 European credit card transactions, each characterized by 30 independent features that have been anonymized using principal component analysis (PCA). The transactions fall between September 1st and September 30th, 2013. A key factor that makes this dataset unique is that it provides a rare, publicly accessible snapshot of real-world credit card usage and patterns, making it the only publicly available dataset of its kind. In the domain of fraud detection, obtaining accurate class labels is particularly challenging and privacy concerns associated with financial data add another layer of complexity. Detecting fraud often requires manual investigation by financial experts, which is often error-prone, expensive, and time-consuming. Even with detailed expert analysis, it is possible for some fraud to remain undetected, giving rise to a need for a robust and automated fraud labeling technique, like the one used in our work.

The credit card dataset is designed for fraud detection, where the provided binary class labels indicate whether a transaction is fraudulent or genuine. It is important to note that these original class labels are used exclusively in evaluating the performance of the models trained for this work. They are not used during the unsupervised feature selection process nor are they used during the unsupervised label generation step. As is typical for fraud detection datasets, the credit card dataset is highly imbalanced, where the examples of fraud are far outnumbered by the genuine. A detailed breakdown of the dataset's composition is

Table 1 Credit card fraud detection dataset class characteristics

Minority count	Majority count	Total count	Minority imbalance	Number of features used
492	284,315	284,807	0.1727%	29/15/10

shown in Table 1. The dataset originally includes 30 independent features, namely *Amount*, *Time*, and *V1* through *V28*, with the *V* features created by the PCA anonymization steps. We chose to omit the Time feature entirely as previous research has shown that it contributes to more noise and is overall not valuable [34]. During the unsupervised SHAP feature selection process, the features are ranked from most important to least important. The top 15 features are, as determined by SHAP, and in order by feature importance: *V13, V26, Amount, V22, V2, V24, V18, V19, V12, V3, V25, V27, V10, V16, and V4*. Thus, indicators for fraudulent credit card use must be more apparent in these features. For the experiments that only use 10 features, we simply choose the top 10 features, starting with *V13*.

Measuring label quality

To effectively evaluate and validate the effectiveness of the newly generated labels, with and without using SHAP feature selection, for fraud detection, we generate new labels across various levels of P and train various supervised classifiers on the labels and measure their classification performance. As P varies, the number of instances labeled as positive, or fraudulent, changes accordingly. However, the total number of instances remains constant within each dataset. The labeling technique is applied to the dataset after the SHAP feature selection step, producing two datasets with different number of features, see Table 1. No instances are omitted during these steps.

It is important to note that the purpose of using the generated class labels with supervised learners is solely to evaluate the performance of the labels themselves, and how unsupervised feature selection affects their quality. We exclude supervised learning with the original labels from our work, as our focus is on scenarios where the data is initially unlabeled. Therefore, using the original labeled data for supervised learning falls outside the scope of this paper.

The classification performance of the supervised models trained on the new labels serves as a valuable indicator of the overall quality of the labels since it mirrors how the labels would be used in practice. In real-world applications, datasets are often unlabeled by default and need to be labeled before use with supervised classifiers. Given the unlabeled dataset, our method would perform feature selection in an unsupervised fashion, and then be used to generate new class labels given just the features as an input. Subsequently, the new labels would be used to train supervised learners to produce classification predictions. The performance of the learners on the new labels is indicative of the expected classification performance on future, unseen fraud detection data. Since we focus on the unlabeled scenario, in the context of our work, a researcher would have two options when using unlabeled data, use the labeling technique to label the dataset and use a supervised learner, or use an unsupervised learner directly with the unlabeled data. It is then crucial to compare the classification performance against that of a baseline approach that does not rely on labels, such as an unsupervised anomaly detection method.

Performance metrics

We use the area under the precision-recall curve (AUPRC) as our primary performance metric, introduced by Provost et al. [35]. AUPRC values are derived from the true positive (TP), false positive (FP), false negative (FN), and true negative (TN) counts as

organized by the classic confusion matrix. The AUPRC effectively summarizes the classification performance trade-off between a model's precision and recall, where precision would be the fraction of correctly identified fraudulent instances out of all instances predicted as fraud, and recall would be the fraction of actual fraudulent instances that were correctly identified as fraud. AUPRC is related to the receiver operating characteristic (ROC) curve, where the true positive rate and false positive rates are plotted against each other, illustrating the trade-off between correctly identified fraud instances and incorrectly classified non-fraud instances. However, we only use AUPRC in part since Davis et al. [36] have shown that ROC curves, and by extension the area under the ROC curve (AUC-ROC or simply AUC) can produce overly optimistic results when dealing with imbalanced datasets. Further, they state that a curve can only dominate in the ROC space if and only if it dominates in the precision-recall (PR) space. Additionally, other studies have shown that AUPRC is a superior metric when working with highly imbalanced fraudulent datasets [34, 37, 38]. The absolute AUPRC score ranges between 0.0 and 1.0, with 1.0 indicating a classifier perfectly identifies all true positives with no misclassifications. That is, the higher the AUPRC, the higher the performance of what is being measured. The minimum AUPRC score is 0.0, but in practice, the lower bound of AUPRC scores corresponds to the class imbalance of the dataset. For example, the dataset we use has a class imbalance ratio of 0.1727%. Thus, the lowest AUPRC score, of a minimally performing model, would be approximately 0.001727.

Classifiers

In this study, we utilize a combination of four supervised learning algorithms, Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), and a Multi-Layer Perceptron (MLP) to evaluate the quality and efficiency of the generated class labels with and without using SHAP feature selection. The dataset chosen is a class-imbalanced fraud detection dataset, which presents unique challenges for labeling and modeling. By training the learners on the generated labels made for this type of dataset and by comparing the performance of these models, we aim to provide an effective measure of the labels themselves, and the overall impact of label quality on model classification performance. That is, we aim to measure the quality of the labeling techniques with feature selection, not to improve supervised learning. This is why we do not perform any hyperparameter tuning of the learners. However, we compare the performance of the supervised models, trained on generated labels, with that of an unsupervised baseline. The baseline must be unsupervised since we focus solely on the unlabeled scenario. Isolation Forest (IF) is used as this baseline as it can be used as an anomaly detection method, and thereby produces a classification performance on a test set, without needing class labels during training.

DT is the first learner we use to measure the effects of feature selection on the generated labels. The DT algorithm is a non-parametric, supervised method known for its interoperability, straightforward implementation, and is widely used. The DT algorithm generates a tree-like structure that consists of a root node, branches, internal nodes, and leaf nodes. Each leaf node represents a class label, and each internal decision node is a decision point based on the data attributes. The DT algorithm operates by recursively partitioning the dataset based on the values of the features, using a

greedy search and a divide and conquer strategy. This process is repeated recursively until most, or all, of the instances have been assigned under a label. As the decision trees grow in size through the training process, its propensity to overfit grows too. To combat this, pruning is used to reduce the size and complexity of the trees and can help avoid overfitting to the training data. Pruning is where branches are removed from the tree if they have low importance. A large advantage of DT is its relative architectural simplicity, allowing them to be more easily interpreted and visualized. Decision trees can be trained quickly with minimal computational requirements. This is in contrast to other, more complex, models such as neural networks. These are often referred to as black box models since they are much harder to interpret or visualize and can require significant computational resources and training time. However, DT's simplicity can be a disadvantage, and they can underperform other models with similar data.

The second supervised classifier used for label evaluation is RF [39]. RF is a widely used and well-known supervised model. It is an ensemble method that is comprised of multiple DTs that are built through bootstrap sampling of the training data, i.e., the data is sampled with replacement. This approach encourages diversity amongst the individual trees, which attempts to reduce the drawbacks of an individual DT. One key feature of RF is that, at each split within a tree, a random subset of the features is considered. In contrast, DT utilizes all features. This results in many trees, each of which focuses on a different, though potentially overlapping, subset of the data, which can produce a model that is more robust against overfitting. RF is well suited for datasets with high dimensionality since it does not need prior feature selection for training. Further, RF maintains the same level interpretability that is provided by the ensemble of decision trees, while also offering a measure of feature importance that identifies the more influential features for classification. The robustness of RF comes from the number of trees; however, the computational and memory requirements increase as the number of trees increases. For classification prediction, RF uses a majority voting mechanism to produce a final class output where each of the member trees contributes its own class prediction.

The third supervised classifier used is LR, also a widely known and commonly used statistical model primarily used for binary classification. The LR model aims to estimate the probability that an input instance belongs to a class by fitting the data to a logistic or sigmoid function. Although the dependent variable is categorical, the relationship between the input features and the dependent variable is non-linear, which can be modeled using log-odds. The sigmoid function is an S-shaped curve that maps any real number to a value between 0 and 1, making it useful for estimating probabilities. To calculate an input's class membership, LR takes a linear combination of the input features and applies the logistic function. A threshold, typically set to 0.5, is used to convert a probability output to a classification output. With respect to binary classification, if the probability prediction is above the threshold, the prediction belongs to one class, if it is below, it is the other class. LR has similar advantages to DT, in that it is relatively simple, interpretable, and straightforward to implement. The coefficients of a trained LR model offer valuable insights into feature importance. LR is a foundational model that has been extensively used across domains, maintaining its relevance in machine learning applications.

The fourth and final supervised classifier we use to measure the quality of the labels is an MLP [40]. The MLP is a type of artificial neural network that is made up of at least three layers: an input layer, a minimum of one hidden layer, and an output layer. It represents the most fundamental architecture for neural networks. Each node in the network, referred to as a neuron or perceptron, in one layer is connected to nodes in the next layer. In a fully connected network, like the one we use, all neurons in one layer are connected to all the neurons in the next. The data flows in one direction from the input layer to the output layer, making these a feedforward network. MLPs are capable of handling non-linearly separable data, making them well-suited for machine learning tasks ranging from the basic to the complex. The neurons in the hidden layers use non-linear activation functions, such as the sigmoid function or rectified linear unit (ReLU) function to their inputs. The connections between the neurons are numeric weights that are incrementally adjusted throughout the training process. Various optimization algorithms can be used during training which specify exactly how the weights are adjusted. For our work, we use backpropagation with gradient descent to train our MLP on the training data and generated labels. Backpropagation calculates the gradient of the loss function for a given input–output pair. These gradients are then used to update the weights to minimize the loss as defined by the loss function. Once this process uses all the training data, an epoch is completed. It is common to train for multiple epochs. The classification performance of an MLP largely depends on the size of the network. However, if the neural network contains too many neurons or layers, it can lead to overfitting.

To thoroughly evaluate the performance of the supervised classifiers, we compare their classification performance against an unsupervised anomaly detection method, used as a baseline. This baseline serves to represent the expected performance in detecting fraud when learning from new, unlabeled data in practice. For this purpose, we use a widely recognized outlier detection method, namely Isolation Forest (IF), originally introduced by Liu et al. in [16].

Like some of our supervised models, IF is a tree-based algorithm. It is designed specifically to isolate anomalies within a given dataset. It does this by recursively partitioning the training data into progressively smaller subsets. An important premise behind IF is that anomalies, by definition, are "few and different" [16], making anomalies more likely to be isolated by IF than normal instances. During the recursion, randomly induced trees are generated until all data points are isolated. In the IF algorithm, anomalies tend to have shorter paths within the trees as compared to the paths of normal data points. This is due to two primary reasons: (1) the inherent class imbalance means anomalies are fewer in number, and (2) instances with separable attributes are more likely to be isolated earlier in the recursion as opposed to later. Since there are many trees within the forest, instances that have consistently shorter paths across many trees are more likely to be labeled anomalous. IF is an appropriate baseline for comparison because it is widely used and has demonstrated better performance than other methods such as Random Forest and Local Outlier Factor on highly imbalanced fraud datasets [41]. For our purposes, when IF identifies an instance as anomalous, it is equivalent to classifying that instance as belonging to the minority class or labeling it as fraud. In our experiments, we train the IF and calculate the classification performance using the data in the test fold. We do not use the generated labels in any way when using IF. Additionally, the IF we use

as a baseline for measuring label quality is a separately trained IF that was used during the SHAP feature selection process.

For all four supervised classifiers and the unsupervised IF, we utilize the implementations provided in Scikit-learn [42], version 1.3.0. When using the DT, we use the Gini criterion for impurity, best splits were chosen over random splits, and all other settings as library defaults. We did not set a maximum tree depth when using DT, allowing the nodes to expand until all leaf nodes are pure or contain fewer than 2 samples but having a minimum of 1 sample. The minimum number of samples required for a split is set to 2. All available features of a given dataset are considered when determining the best split. When looking at the two datasets generated by SHAP feature selection, the DT is limited to the number of features for a given dataset: in our experiments we use all (29), 15, and 10 features. For RF, we use 100 trees in the forest, and similar settings to DT: Gini criterion, minimum of 2 samples for a split, and a minimum of 1 sample per leaf node. However, unlike the DT, we set the maximum tree depth to 4 for our RF. When using LR, we set all parameters to the library defaults. The architecture for the MLP consists of a single hidden layer with 100 neurons, ReLU activation function, and the Adam optimization algorithm, which is a variant of stochastic gradient descent [43]. Log-loss is used as the MLP's loss function. We train using a mini-batch size of 200, constant learning rate, and for 300 epochs, and default settings for early stopping.

There is only one parameter for IF [16], called the contamination rate, and it is left as the library default. However, during our experiments and experience with IF, we observed that the contamination rate parameter did not affect any threshold-based performance metrics, such as AUPRC. It only affects performance metrics that use classification values themselves, such as accuracy, as opposed to a threshold-based metric. As a note, any other parameters not stated thus far were left as library defaults for ease of comparison. We aim to evaluate the quality of the labels and the effects of unsupervised feature selection on the generated labels across several different types of supervised models, not to find the best configuration for a given model. Thus, hyperparameter tuning for the classifiers is out of scope of this paper and will not provide any additional insights in the quality of the labels themselves. For the experiments, we use 5-fold cross validation, where the training splits use the generated labels, and the test splits use the original labels provided by the dataset. We repeat this for 10 rounds to generate enough data samples for our statistical analysis. All the AUPRC scores shown in the tables, unless otherwise noted, are the average value across the 50 trials.

Experimental setup

We first apply the SHAP feature selection technique to the credit card fraud detection dataset to measure the feature importance of the datasets, in an unsupervised fashion. For our experiments SHAP was used to produce three credit card datasets, the original one with all 29 features, one with the top 15 features, and one with the top 10 features. Next, we apply the unsupervised labeling technique to each of these three datasets so we can effectively measure the effects of unsupervised feature selection with newly generated class labels for credit card fraud detection.

Next, for each of the three levels of feature counts, we set the method's inputs to produce a labeled dataset with P = 500, 1000, and 1500. During experimentation for this

work and previous work [12], we observed the performance to reach a plateau near P = 1500. The initial value of 500 for P was chosen to model a practitioner making the initial estimate that the class ratio of the data, suspected to be highly imbalanced, to be approximately 0.002. A decision that, in practice, would be guided by domain expertise. A total of 9 datasets were created for our experiments, each of which had the 4 supervised learners trained with 5-fold cross validation for 10 replications, and an Isolation Forest trained with 5-fold cross validation with 10 replications, resulting in 2250 experimental outcomes.

Although the SHAP feature selection technique and the labeling technique exclusively uses the dataset features to generate feature importance and class labels, the original class labels provided by the dataset are preserved for each instance. These original labels are solely used to calculate the AUPRC score for the supervised classifiers and unsupervised IF. This process is integrated into the 5-fold cross-validation steps where there is an 80/20 train/test split for each fold. The models were trained on the 80% splits, which had the generated labels, and tested on the 20% split that had the ground truth labels. We apply the stratified K-fold method, which strives to maintain the class imbalance ratio in the splits. Though the models only use the generated label in the training folds, the ground truth labels are used for the splits. This approach is crucial because it ensured that the data in the test splits reflects a similar distribution and characteristic as the overall dataset, providing test data that is representative of the domain. The same 5-fold cross validation setup was used when working with the IF model. However, since IF is unsupervised, the generated labels are not used in training. The ground truth labels are used in the test split to generate an AUPRC that we use to compare to the other models.

Statistical analysis

We use two-factor analysis of variance (ANOVA) to test for statistical differences between our experimental outcomes. The average AUPRC results across each 50 experimental replications, for each number of P and number of features selected, is shown in Table 3. We assess whether the mean AUPRC scores vary significantly when using a different number of features and across the three levels of P. As shown in Table 2, the

Table 2 ANOVA

Model	Factor	DF	Sum Sq	Mean SQ	F Value	Pr (> F)
DT	Features	2	0.1863	0.09316	88.38	$< 2 \times 10^{-16}$
	P	2	0.2435	0.12175	115.5	$< 2 \times 10^{-16}$
	Residuals	445	0.4691	0.00105		
RF	Features	2	0.7303	0.3652	168.385	$< 2 \times 10^{-16}$
	P	2	0.007	0.0035	1.619	0.199
	Residuals	445	0.965	0.0022		
LR	Features	2	0.445	0.22251	96.739	$< 2 \times 10^{-16}$
	P	2	0.0016	0.00082	0.355	0.701
	Residuals	445	1.0235	0.0023		
MLP	Features	2	0.3464	0.17321	90.52	$< 2 \times 10^{-16}$
	P	2	0.227	0.1135	59.31	$< 2 \times 10^{-16}$
	Residuals	445	0.8515	0.00191		

Table 3 Average classifier AUPRC performance results with respect to number of features and number of positive-labeled instances *P*

Features	P	DT	RF	LR	MLP
10	500	0.3304	0.2966	0.2231	0.2415
	1000	0.3788	0.2901	0.2131	0.2993
	1500	0.3912	0.2757	0.1805	0.2953
15	500	0.3569	0.3703	0.2968	0.2816
	1000	0.3761	0.3853	0.2623	0.3059
	1500	0.3890	0.3985	0.2378	0.3044
29	500	0.2935	0.3394	0.2301	0.1904
	1000	0.3181	0.3529	0.2793	0.2345
	1500	0.3714	0.3596	0.3226	0.2692

number of features selected, denoted in the table as simply "Features", has an effect on the AUPRC performance on all four learners. This is denoted by a very small p-value of less than 2×10^{-16} in all four cases. This allows us to reject the null hypothesis and conclude that there are statistically significant differences between AUPRC values depending on the number of features selected. The results also show that for DT and MLP, the number of positive instances P has an effect on the AUPRC result, but it does not have effect on AUPRC when using RF or LR. Since we only use supervised learners to measure label quality, this implies that the model itself influences the label quality measurement. Future work includes exploring additional ways of measuring generated label quality.

To further analyze these differences, we perform Tukey's Honestly Significant Difference (HSD) test, which is a widely used post-ANOVA pairwise comparison test to determine which groups differ from each other [44, 45]. However, for the Tukey's HSD test, we use the average AUPRC value across all levels of P because we are interested in the effects of unsupervised feature selection on generated label quality. Providing an in-depth statistical analysis across different P values is out of scope since our primary focus of this paper is to analyze how unsupervised feature selection effects AUPRC performance of the labels, rather than an in-depth analysis of the number of positive-labeled instances. However, we do present the results in Table 3 and briefly discuss them. Additionally, we do not directly compare the different supervised models between each other with ANOVA or a Tukey's HSD test because the supervised models are only used to measure the quality of the labels. Directly comparing the supervised models would suggest that there is an optimal supervised model related to the unsupervised labeling and feature selection techniques. This is not the case, and the presentation of results obtained from different supervised classifiers is only intended to illustrate the differences in expected performance, depending on what model is using the labels, across different features selected. We use the Tukey's HSD test to rank the AUPRC performance across the three levels of feature selection for each of the four supervised models, DT, RF, LR, and MLP. Within each of those four, we also rank three different IF AUPRC performances. IF-10, IF-15, and IF-29 are the results from using IF on the dataset that used the top 10, 15, and all 29 features. The Tukey's HSD test results are shown in Table 5. For clarity, the IF AUPRC results are shown in their own table, shown in Table 4.

Table 4 Isolation forest performance results

Features used	AUPRC
10	0.2160
15	0.2630
29	0.2434

Table 5 Credit Card Fraud Tukey's HSD Tests

DT	Features	15	10	29	IF-15	IF-29	IF-10
	AUPRC	0.3740	0.3668	0.3277	0.2630	0.2434	0.2160
	Group	a	a	b	c	c	d
RF	Features	15	29	10	IF-15	IF-29	IF-10
	AUPRC	0.3847	0.3506	0.2875	0.2630	0.2434	0.2160
	Group	a	b	c	d	d	e
LR	Features	29	15	IF-15	IF-29	IF-10	10
	AUPRC	0.2773	0.2657	0.2630	0.2434	0.2160	0.2056
	Group	a	a	ab	b	c	c
MLP	Features	15	10	IF-15	IF-29	29	IF-10
	AUPRC	0.2973	0.2787	0.2630	0.2434	0.2314	0.2160
	Group	a	b	bc	cd	de	e

The outcome of a Tukey's test on a pair of groups provides a range of confidence intervals. If the interval includes zero, it suggests there is no significant difference between the two groups. For example, if an AUPRC result that belongs to group "a" and another result belongs to group "b", that means they are statistically different, and the one with a higher AUPRC value would be better with a statistical significance. However, if both AUPRC results fall into the same group (e.g. both are in "a"), it means that there is no statistically significant difference in performance, but one mean may be higher than the other. If multiple results share grouping letters, groups with the same letter do not differ statistically. For example, if three AUPRC results are in groups "a", "ab", and "b", the first and third ranked groups are statistically significantly different, but the second group is statistically similar to the first and the third group.

Results and analysis

We present the AUPRC results for DT, RF, LR, and MLP across three levels of P and across the three levels of feature selection in Table 3. When evaluating the label quality using the DT model, the AUPRC classification performance steadily increases from 0.3304 to 0.3912 as P goes from 500 to 1500 when using the top 10 features. This trend, of increasing AUPRC, can be observed for all three levels of features selected. Though the performance between the number of features differs. The same general pattern can be observed for the MLP learner. The AUPRC increases as P increases when using all 29 features. However, the AUPRC performance only increases when P goes from 500 to 1000 when using the top 10 and 15 features. The ANOVA table, Table 2, shows that for both DT and MLP, the number positive-labeled has an impact for these learners, with low p-values. In contrast, the AUPRC scores for RF and LR are a bit different. With these two learners, the p-value shows that there is no statistical impact from P. However, the

average AUPRC scores shown in Table 3 show that the average changes as P changes. From these two observations we can conclude that there is a large variance in AUPRC score across the 50 experimental trials for these. Though, for the DT and MLP, the learners with small p-values, the AUPRC steadily increases with P.

Next, we examine the AUPRC results for the four supervised leaners, averaged across the three P values, and compare them across the levels of feature selection, shown in Table 5. Table 5 is sorted from highest to lowest AUPRC, from left to right, showing which group each set of results belongs to. Additionally, the IF results are repeated in each row. This is done to easily compare the supervised classifier results to the IF results, on a per-classifier basis. We also compare the results of IF across the same levels of feature selection. Table 5 shows the results of a Tukey's HSD test and the resulting rankings of the AUPRC score. The results for DT show that using generated labels outperforms IF on all accounts. The IF results, IF-15, IF-29, and IF-10 are IF applied to the dataset with the top 15, all, and top 10 features, respectively. All the IF results are in uniquely lower HSD groups than all the DT results. The unsupervised SHAP feature selection improves the quality of the generated labels when looking at the AUPRC results for DT. Though the highest average AUPRC result for DT is when the top 15 features are used, the HSD groupings show that using either 10 or 15 features produces statistically similar results, as denoted by both being in the "a" group. Using all 29 features, has a lower AUPRC and is in a different group. This shows that unsupervised feature selection significantly improves the AUPRC performance of the labels when using DT to measure the quality.

When using RF to measure the quality of the labels, the generated labels always outperform that of IF, similar to DT. With these results, each of the 6 results are in distinct HSD groups. Like with the DT results, the RF results show that using some level of feature selection produces the highest AUPRC and that the right level of feature selection improves the quality of the generated labels. In this case, using the top 15 features produces the best labels, as measured by the AUPRC of RF. RF achieved an AUPRC of 0.3847, the highest AUPRC score we measured. These results also show that using the top 10 features decreases the quality of the labels when using RF. This contrasts with DT, where using the top 10 makes the quality of labels such that it is better than not using feature selection when using DT as a measure. The results for IF are also consistent with the top 15, or all features producing statistically similar results, as can be seen with the groups of "c" and "d" in DT and RF, respectively. This shows that SHAP unsupervised feature selection produces a net positive result, even when just using the IF baseline. Though these results underperform that of the generated label results.

The AUPRC results for LR in Table 5 show a slightly different pattern than the other three. Here, we show that not using feature selection and using the top 15 features produce statistically similar results, as denoted by both of the being in group "a". These results though, still outperform the IF and when using the top 10 features. Though there is no AUPRC advantage to using feature selection, it still would be a better choice to use fewer features, rather than more, given the same performance. This is due to the model's reduced complexity and faster runtime when using fewer features. With fewer input variables, a model becomes simpler, which reduces overfitting and can improve its ability to generalize to new data. Additionally, training and prediction times are reduced with fewer features to process. Furthermore, using fewer features minimizes the risk of

multicollinearity, leading to more stable and reliable performance. The AUPRC results from MLP show more similarity with that of DT and RF. They show that using unsupervised feature selection improves the quality of the generated labels, regardless of how many features selected, and these outperform all of the IF results. Across all four learners, using unsupervised SHAP feature selection, and using the top 15 features, significantly improves the quality of the generated labels and outperforms that of IF. It also shows that feature selection can help IF performance.

Conclusion

Large datasets are often unlabeled, initially. Labeling a dataset typically requires manual human intervention that comes at substantial financial and time costs, as well as the risks associated with inaccurate or noisy labels. Financial fraud detection data, such as the credit card fraud detection dataset used in our study, have even more challenges, namely data privacy and high-class imbalance. Both of these present challenges to machine learning, but the latter, in particular, can severely diminish classification performance of learners using such data.

In this study, we present a detailed evaluation of the quality improvements an unsupervised feature selection technique can provide to generated class labels for highly imbalanced credit card fraud detection data. The SHAP feature selection method employed is capable of ranking feature importance without the need for class labels. Once the features are ranked, three different datasets are created with different number of features. These datasets then have class labels generated by the labeling technique. These approaches effectively address the challenges of large unlabeled data as well as challenges posed by high class imbalance, while reducing the required number of features, all in a completely unsupervised manner.

We measure the quality of the labels by training a variety of supervised classifiers on the generated labels and compare their AUPRC results to that of an unsupervised baseline. Our empirical results show that, for the credit card transaction data used in this study, in combination with an unsupervised labeling technique, unsupervised feature selection significantly improves the quality of the generated labels. This improvement is evident when comparing to generated labels created using all the features, as well as compared to the baseline. Feature selection consistently improves the quality of the generated fraud labels. Results show that using the top 15 features, as selected by SHAP, is the highest performer, and produces the highest quality labels. We can conclude that the SHAP feature selection technique improves label quality in all but one example. In that case, SHAP feature selection performs similarly to using all features, but would be the better choice since using a reduced set of features, to achieve the same AUPRC, is ideal due to the reduced size and complexity of the model. Future work includes exploring other unsupervised feature selection techniques and evaluating other large highly imbalanced datasets in other domains.

Abbreviations

ANOVA	Analysis of variance
AUPRC	Area under the precision-recall curve
DT	Decision tree
FN	False negative
FP	False positive

FPR	False positive rate
HSD	Tukey's honestly significant difference
IF	Isolation forest
LR	Logistic regression
MLP	Multilayer perceptron
MSE	Mean squared error
NN	Neural network
PR	Precision-recall
ReLu	Rectified linear unit
RF	Random forest
ROC	Receiver operating characteristic curve
SHAP	SHapley Additive exPlanations
TN	True negative
TP	True positive
TPR	True positive rate

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RKLK carried out the concept and design of the research, performed the implementation and experimentation, and contributed to the manuscript. TMK contributed to the design of the research. TMK and FV provided reviews to the manuscript. All authors read and approved the final manuscript.

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References

1. Deng J, Dong W, Socher R, Li L.-J, Li K, Fei-Fei L. Imagenet: a large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition, pp. 248–255 (2009). Ieee
2. Zhang C, Bengio S, Hardt M, Recht B, Vinyals O. Understanding deep learning (still) requires rethinking generalization. Commun ACM. 2021;64(3):107–15.
3. Halevy A, Norvig P, Pereira F. The unreasonable effectiveness of data. IEEE Intell Syst. 2009;24(2):8–12.
4. Sun C, Shrivastava A, Singh S, Gupta A. Revisiting unreasonable effectiveness of data in deep learning era. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 843–852. 2017.
5. Litjens G, Kooi T, Bejnordi BE, Setio A, Ciompi F, Ghafoorian M, Van Der Laak JA, Van Ginneken B, Sánchez CI. A survey on deep learning in medical image analysis. Med Image Anal. 2017;42:60–88.
6. Debener J, Heinke V, Kriebel J. Detecting insurance fraud using supervised and unsupervised machine learning. J Risk Insurance. 2023;90(3):743–68.
7. Xie J, Girshick R, Farhadi A. Unsupervised deep embedding for clustering analysis. In: International Conference on Machine Learning, pp. 478–487. 2016. PMLR.
8. Mingers J. An empirical comparison of pruning methods for decision tree induction. Mach Learn. 1989;4:227–43.
9. Srivastava N, Hinton G, Krizhevsky A, Sutskever I, Salakhutdinov R. Dropout: a simple way to prevent neural networks from overfitting. J Mach Learn Res. 2014;15(1):1929–58.
10. Krawczyk B. Learning from imbalanced data: open challenges and future directions. Prog Artif Intell. 2016;5(4):221–32.
11. Khoshgoftaar TM, Seiffert C, Van Hulse J, Napolitano A, Folleco A. Learning with limited minority class data. In: Machine Learning and Applications, 2007. ICMLA 2007. Sixth International Conference On, pp. 348–353. 2007. IEEE.
12. Kennedy RK, Villanustre F, Khoshgoftaar TM, Salekshahrezaee Z. Synthesizing class labels for highly imbalanced credit card fraud detection data. J Big Data. 2024;11(1):1–22.

13. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In: Guyon I, Luxburg UV, Bengio S, Wallach H, Fergus R, Vishwanathan S, Garnett R. (eds.) Advances in Neural Information Processing Systems 30, pp. 4765–4774. Curran Associates, Inc., Red Hook, NY, USA (2017). http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf.
14. Dal Pozzolo A, Caelen O, Johnson RA, Bontempi G. Calibrating probability with undersampling for unbalanced classification. In: 2015 IEEE Symposium Series on Computational Intelligence, pp. 159–166. 2015. IEEE
15. Kaggle: Credit Card Fraud Detection. https://www.kaggle.com/mlg-ulb/creditcardfraud. 2018.
16. Liu FT, Ting KM, Zhou Z-H. Isolation forest. In: 2008 Eighth Ieee International Conference on Data Mining, pp. 413–422. 2008. IEEE.
17. Leevy JL, Hancock J, Khoshgoftaar TM, Abdollah Zadeh A. Investigating the effectiveness of one-class and binary classification for fraud detection. J Big Data. 2023.
18. Hancock JT, Khoshgoftaar TM, Liang Q. A problem-agnostic approach to feature selection and analysis using shap. J Big Data. 2024. https://doi.org/10.1186/s40537-024-01041-1.
19. Salekshahrezaee Z, Leevy JL, Khoshgoftaar TM. The effect of feature extraction and data sampling on credit card fraud detection. J Big Data. 2023;10(6):1–17.
20. Masci J, Meier U, Cireşan D, Schmidhuber J. Stacked convolutional auto-encoders for hierarchical feature extraction. In: Artificial Neural Networks and Machine Learning–ICANN 2011: 21st International Conference on Artificial Neural Networks, Espoo, Finland, June 14-17, 2011, Proceedings, Part I 21, pp. 52–59. 2011. Springer.
21. Jiang S, Dong R, Wang J, Xia M. Credit card fraud detection based on unsupervised attentional anomaly detection network. Systems. 2023;11(6):305.
22. Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y. Generative adversarial nets. Adv Neural Inf Proc Syst. 2014;27.
23. Rezapour M. Anomaly detection using unsupervised methods: credit card fraud case study. Int J Adv Comput Sci Appl. 2019;10(11).
24. Schölkopf B, Williamson R.C, Smola A, Shawe-Taylor J, Platt J. Support vector method for novelty detection. Adv Neural Inf Proc Syst. 1999;12.
25. Ng A, et al. Sparse autoencoder. CS294A Lecture notes. 2011;72(2011): 1–19.
26. Moslehi F, Haeri A, Gholamian MR. A novel selective clustering framework for appropriate labeling of clusters based on k-means algorithm. Scientia Iranica. 2020;27(5):2621–34.
27. Rauber A. Labelsom: on the labeling of self-organizing maps. In: IJCNN'99. International Joint Conference on Neural Networks. Proceedings (Cat. No. 99CH36339), vol. 5, pp. 3527–3532. 1999. IEEE.
28. Kohonen T. The self-organizing map. Proc IEEE. 1990;78(9):1464–80.
29. Maqbool O, Babri HA. Automated software clustering: an insight using cluster labels. J Syst Softw. 2006;79(11):1632–48.
30. Kennedy RK, Salekshahrezaee Z, Khoshgoftaar TM. A novel approach for unsupervised learning of highly-imbalanced data. In: 2022 IEEE 4th International Conference on Cognitive Machine Intelligence (CogMI), pp. 52–58. 2022. IEEE.
31. Wan Z, Zhang Y, He H. Variational autoencoder based synthetic data generation for imbalanced learning. In: 2017 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 1–7. 2017. IEEE.
32. Kennedy RK, Salekshahrezaee Z, Khoshgoftaar TM. Unsupervised anomaly detection of class imbalanced cognition data using an iterative cleaning method. In: 2023 IEEE 24th International Conference on Information Reuse and Integration for Data Science (IRI), pp. 303–308. 2023. IEEE.
33. Chollet F, et al. Keras. https://keras.io. 2015.
34. Leevy JL, Khoshgoftaar TM, Hancock J. Evaluating performance metrics for credit card fraud classification. In: 2022 IEEE 34th International Conference on Tools with Artificial Intelligence (ICTAI), pp. 1336–1341. 2022. IEEE.
35. Provost FJ, Fawcett T, et al. Analysis and visualization of classifier performance: comparison under imprecise class and cost distributions. In: KDD, vol. 97, pp. 43–48. 1997.
36. Davis J, Goadrich M. The relationship between precision-recall and roc curves. In: Proceedings of the 23rd International Conference on Machine Learning, pp. 233–240. 2006.
37. Wang H, Liang Q, Hancock JT, Khoshgoftaar TM. Enhancing credit card fraud detection through a novel ensemble feature selection technique. In: 2023 IEEE 24th International Conference on Information Reuse and Integration for Data Science (IRI), pp. 121–126. 2023. IEEE.
38. Hancock JT, Khoshgoftaar TM, Johnson JM. Evaluating classifier performance with highly imbalanced big data. J Big Data. 2023;10(1):42.
39. Breiman L. Random forests. Mach Learn. 2001;45:5–32.
40. Rosenblatt F. The perceptron: a probabilistic model for information storage and organization in the brain. Psychol Rev. 1958;65(6):386.
41. Bauder RA, da Rosa R, Khoshgoftaar TM. Identifying medicare provider fraud with unsupervised machine learning. In: 2018 IEEE International Conference on Information Reuse and Integration (IRI), pp. 285–292. 2018. IEEE.
42. scikit-learn: Scikit-learn. http://scikit-learn.org/stable/.
43. Kingma DP, Ba J. Adam: a method for stochastic optimization. arXiv preprint arXiv:1412.6980. 2014.
44. Abdi H, Williams LJ. Tukey's honestly significant difference (hsd) test. Encyclopedia of Research Design. Thousand Oaks, CA: Sage, 1–5. 2010.
45. Berenson M, Levine D, Goldstein M. Intermediate statistical methods and applications: a computer package approach. Englewood Cliffs: Prentice-Hall; 1983.

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